

MEMBER 3 — AI/ML ENGINEER BREAKDOWN

Main Goal

Develop the complete AI-powered face recognition system that:

- captures faces
- generates face encodings
- recognizes students
- verifies identity
- sends attendance data to backend

This is the core intelligent module of the entire project.

Since you are taking this role, your part will become the main “innovation” section during presentation and viva.

PHASE 1 — AI Environment Setup

Objective

Setup AI development environment and dependencies.

Technologies

- Python
- OpenCV
- face_recognition
- NumPy
- Pickle

Tasks

Install Required Libraries

opencv-python

face_recognition

numpy
pickle-mixin

Create AI Folder Structure

```
ai_module/  
|  
|—— register_face.py  
|—— recognize_face.py  
|—— encode_faces.py  
|—— utils.py  
|—— dataset/  
|—— encodings/
```

Setup Webcam Access

Features:

- webcam initialization
 - live frame reading
 - frame display
-

Deliverables

- AI environment configured
 - Webcam feed working
-

PHASE 2 — Face Detection System

Objective

Detect human faces from webcam feed.

Tasks

Live Face Detection

System should:

- detect face in real-time
 - draw rectangle around face
 - process webcam frames continuously
-

Multiple Face Handling

If multiple faces appear:

- detect all faces
 - process one-by-one
-

No Face Detection Handling

Display:

No Face Detected

Deliverables

- Real-time face detection system
-

PHASE 3 — Student Face Registration System

Objective

Capture and register student face data.

Tasks

Capture Face Samples

Instead of 1 image:

capture:

- 10–20 frames

This improves recognition accuracy.

Face Quality Validation

Check:

- face visibility
- blur
- lighting
- distance

Reject bad frames.

Generate Face Encodings

Convert detected face into:

- numerical encoding vector

This is the most important AI step.

Save Encodings

Store:

- encoding vectors
- student ID mapping

Use:

- Pickle file
OR
 - database storage
-

Deliverables

- Student registration AI system
-

Objective

Recognize registered students using webcam.

✅ Tasks

Live Recognition Workflow

Webcam Feed



Face Detection



Face Encoding



Compare With Stored Encodings



Identify Student

Recognition Matching

Compare:

- live face encoding
with:
 - stored encodings
-

Display Recognized Student

Example:

Recognized: Rahul Sharma

📦 Deliverables

- Real-time recognition system
-

Objective

Ensure attendance marking is reliable and accurate.

✓ Tasks

Confidence Score System

Calculate:

- recognition confidence percentage

Example:

Confidence: 93%

Recognition Threshold

If confidence:

- above threshold → accept
- below threshold → reject

Example:

if confidence > 80:

mark_attendance()

Unknown Face Detection

If face not matched:

⚠ Unknown Person Detected

This is an excellent presentation feature.

📦 Deliverables

- Reliable attendance verification system
-

Objective

Connect AI module with backend APIs.

Tasks

Send Attendance Data To Backend

When student recognized:

Send:

- student ID
- name
- timestamp

to backend API.

Receive Backend Response

Possible responses:

Success

Attendance Marked

Duplicate

Attendance Already Marked

Unknown

Student Not Found

Deliverables

- AI-backend communication working
-

PHASE 7 — Recognition Optimization

Objective

Improve speed and accuracy.

✅ Tasks

Face Encoding Caching

Instead of:

- reading images repeatedly

Load encodings once into memory.

This improves:

- speed
- performance

Very important viva point.

Frame Optimization

Process:

- every 2nd or 3rd frame

Improves FPS and smoothness.

Resize Frames Before Processing

Example:

```
frame = cv2.resize(frame, (0,0), fx=0.25, fy=0.25)
```

Improves recognition speed.

📦 Deliverables

- Optimized recognition pipeline
-

📌 PHASE 8 — Error Handling & Testing

Objective

Make AI module stable for final demo.

Tasks

Handle Edge Cases

Low Light

Side Face

Multiple Faces

Camera Failure

No Face Detected

Accuracy Testing

Test:

- different lighting
 - different distances
 - different angles
-

Recognition Performance Testing

Measure:

- detection speed
 - accuracy rate
-

Deliverables

- Stable AI module
-

PHASE 9 — Advanced Extra Features

Objective

Add high-value features for better marks.

✓ Recommended Features

✓ Unknown Person Alert

If unknown detected:

- show alert
- optionally log event

Excellent for security demo.

✓ Attendance Cooldown System

Prevent repeated scanning within few seconds.

Example:

```
if scanned_within_30_seconds:  
    ignore_scan()
```

✓ Live Recognition Box

Display:

- student name
- confidence score
above face rectangle.

Very professional-looking demo.

✗ Features Removed

Emotion Detection

Removed because:

- unreliable
- unnecessary complexity
- unrelated to attendance
- risky during live demo

Keeping project stable is more important.

Multiple Camera Monitoring

Too large for semester project scope.

Deliverables

- Full AI attendance system
- Optimized recognition
- Unknown person detection
- Backend integration